

	Work Instruction	Doc. Revision	00
	AXI V810 X-Axis Read Head Encoder Replacement Procedure	Effective Date	25 Jan 2017

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☒	S1	☒	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	25 Jan 2017	First Release	TAN LIANG KAI

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Removal of X-Axis Encoder

1. Part ID for X-Axis Read Head Encoder (Figure 1):
 - a) Standard machine (Part ID: 4XASP-A039)
 - b) S2EX machine (Part ID: 4JASC-A082)
 - c) XXL machine (Part ID: 4XASP-A095)



Figure 1 X-Axis Read Head Encoder

2. Turn off X-ray source and SSCA power.
3. Open left hand side outer barrier
 - a) For standard machine
 - i. Unplug all cables shown in Figure 2.

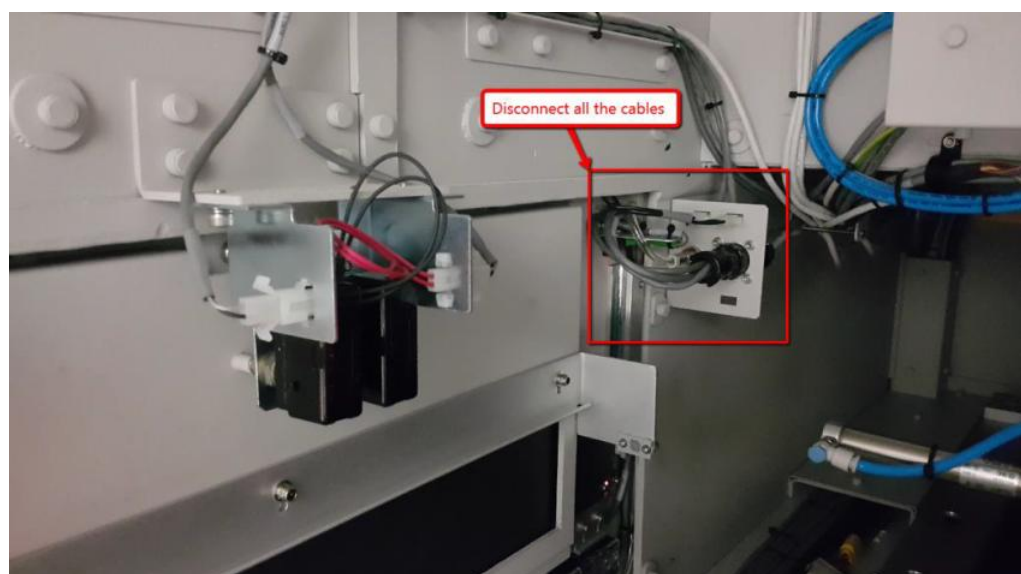


Figure 2

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- ii. Remove 4 screws as shown in Figure 3.

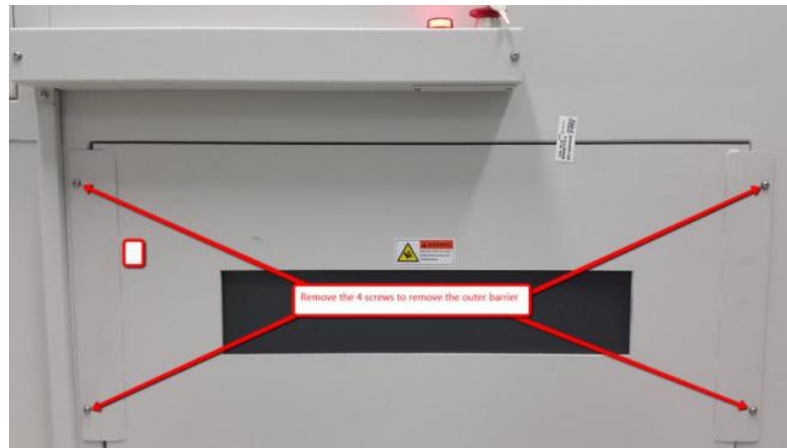


Figure 3

- iii. Gently remove outer barrier from machine.

- b) For S2EX and XXL machine

- i. Remove 2 screws as shown in Figure 4.



Figure 4

- ii. Open outer barrier.

4. Open rear access door.

5. Loosen 2 screws and remove lead cover as shown in Figure 5.

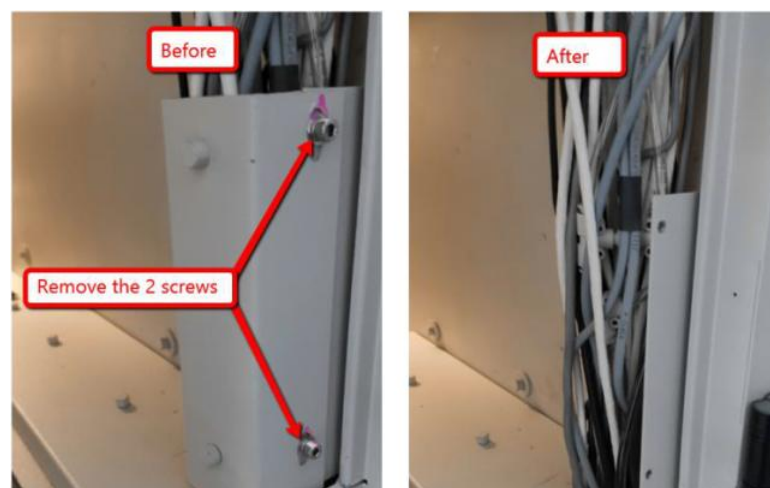


Figure 2

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6. Loosen 3 x screws and remove the lead cover as shown in Figure 6.

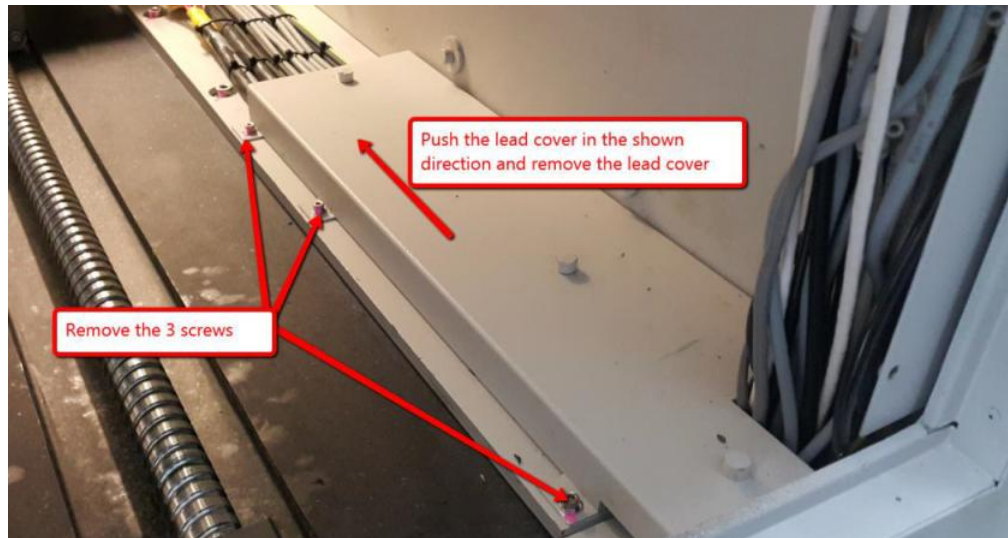


Figure 3

7. Loosen 2 screws and remove bracket as shown in Figure 7.

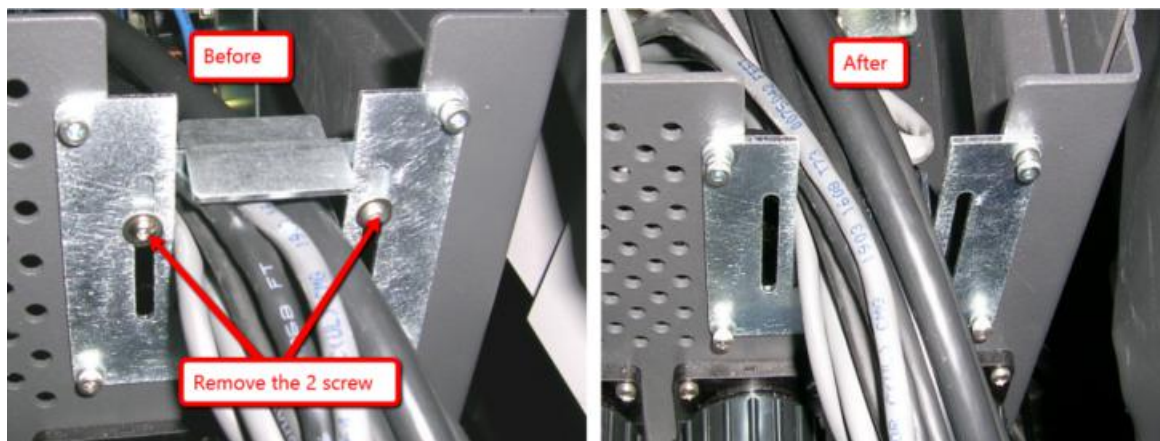


Figure 4

8. Unplug X-Axis encoder cable connector from X-Axis kollmorgen as shown in Figure 8.

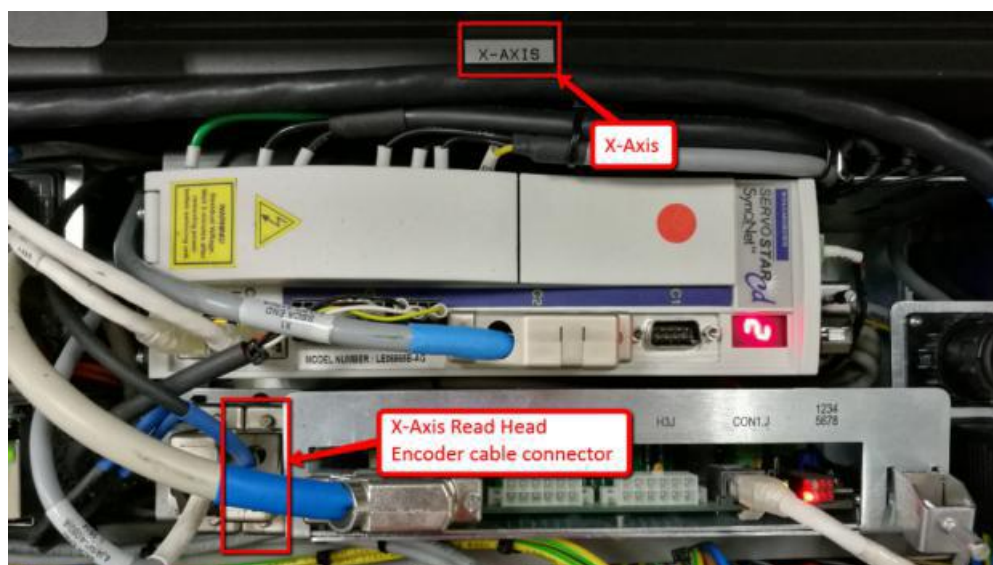


Figure 5

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9. Remove X-Axis encoder cable from SSCA by following cable routing as shown in Figure 6 & 7.
Remark: Remove necessary cable ties and nicely place the encoder cable.

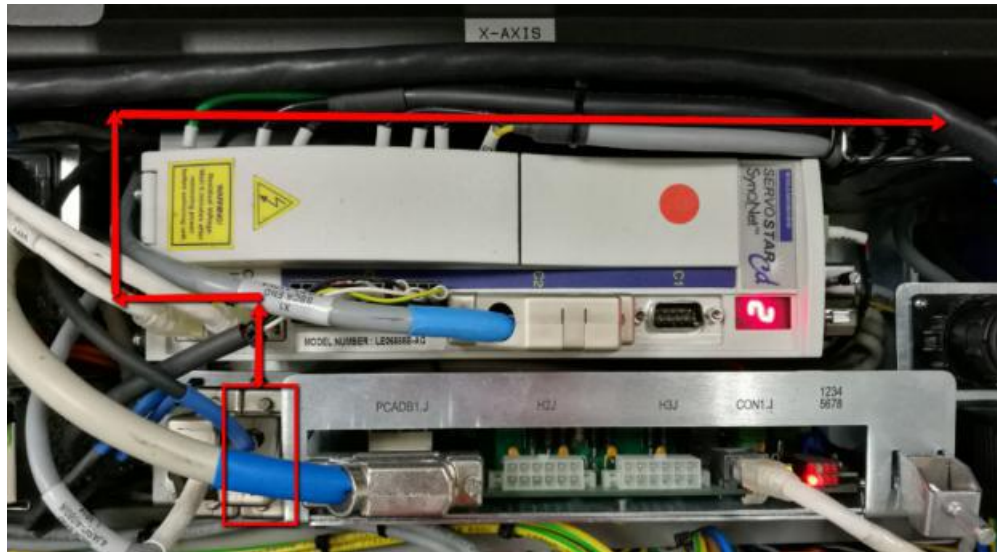


Figure 6

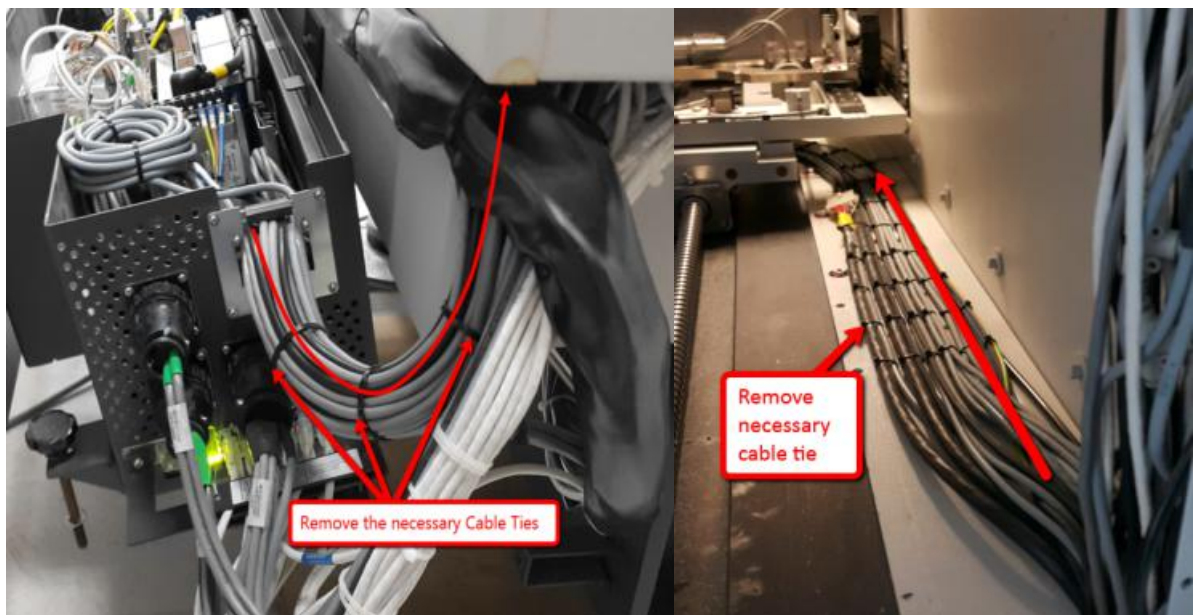


Figure 7

10. Remove necessary cable ties on the bracket as shown in Figure 8.



Figure 8

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11. Unplug X-Axis motor cable connector and loosen the screw as shown in Figure 9.

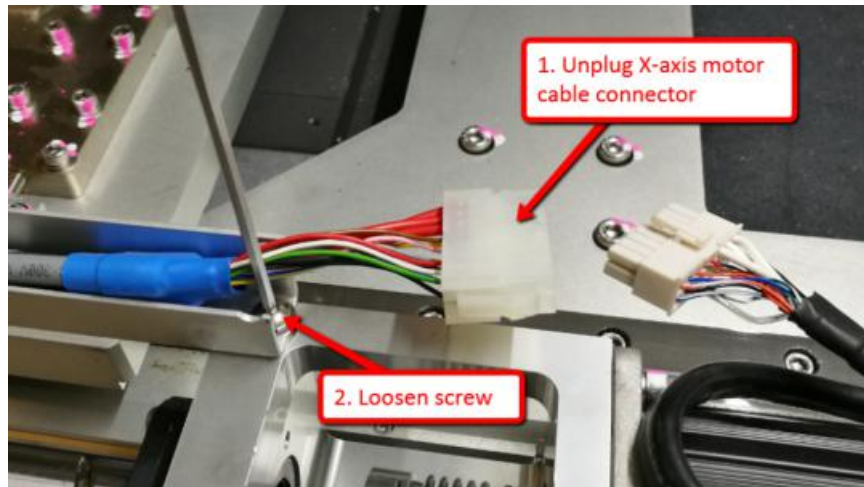


Figure 9

12. Loosen 2 screws and remove X-Axis E-chain Guard plate and X-Axis motor cable bracket as shown in Figure 10.

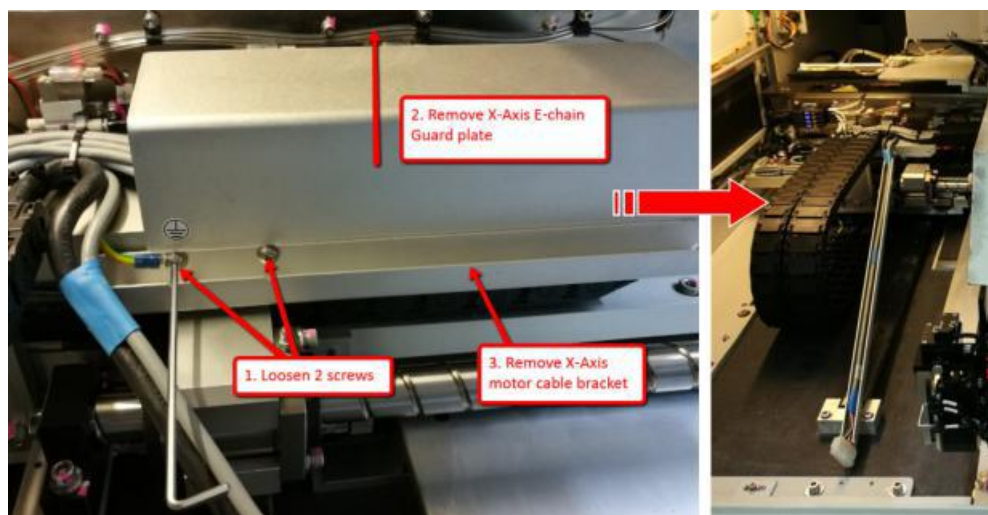


Figure 10

13. Remove 2 screws and cable clamp as shown in Figure 11 below.



Figure 11

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14. Loosen 2 screws by using camera tool and remove X-Axis Encoder as shown in Figure 12.

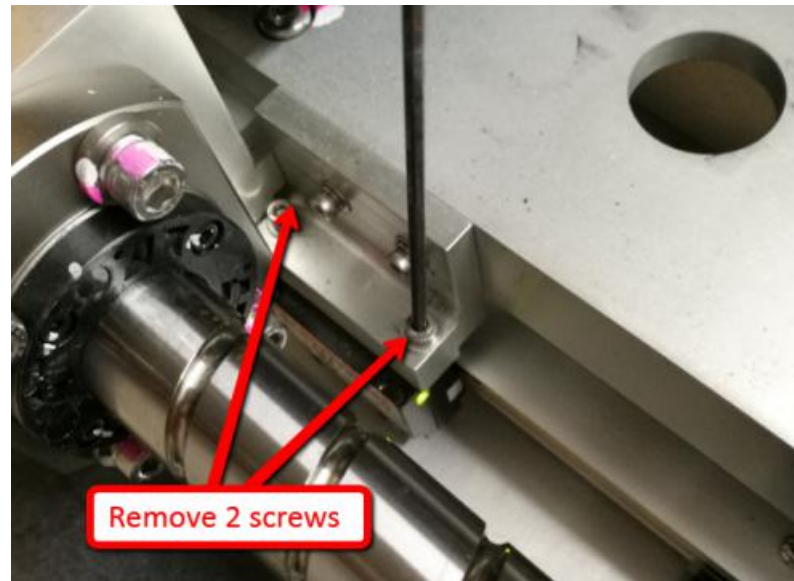


Figure 12

15. Use flat head screw driver to open first X-Axis E-chain slot cover as shown in Figure 13.

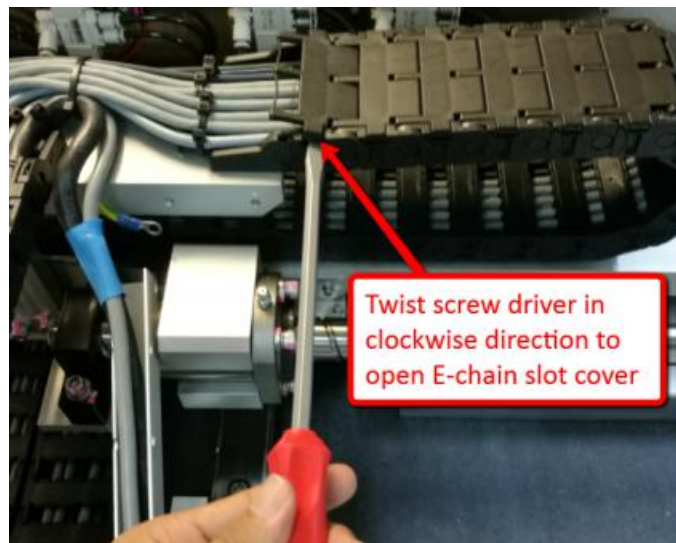


Figure 13

16. Remove necessary cable ties as shown in Figure 14.

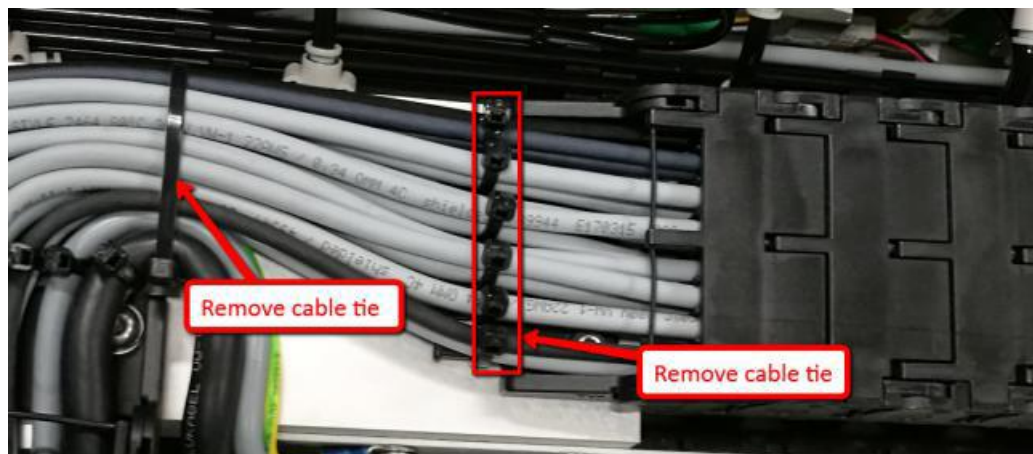


Figure 14

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17. Use flat head screw driver to open X-Axis E-chain top joint as shown in Figure 15.

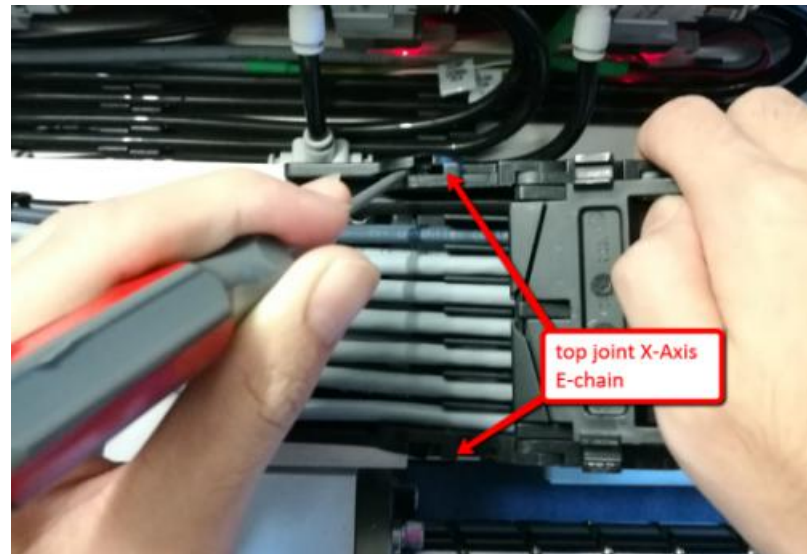


Figure 15

18. Loosen 2 screws and remove encoder cable as shown in Figure 16.

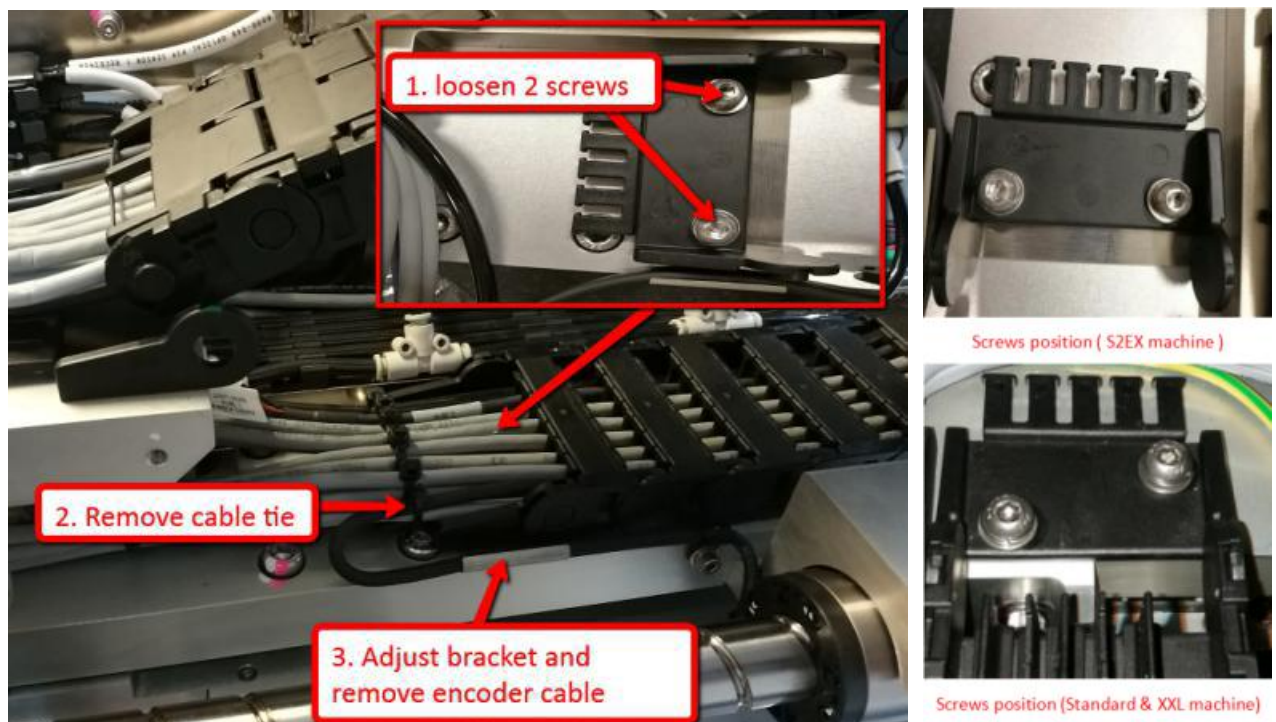


Figure 16

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19. Use flat screw driver to open X-Axis E-chain Bottom joint as shown in Figure 17.

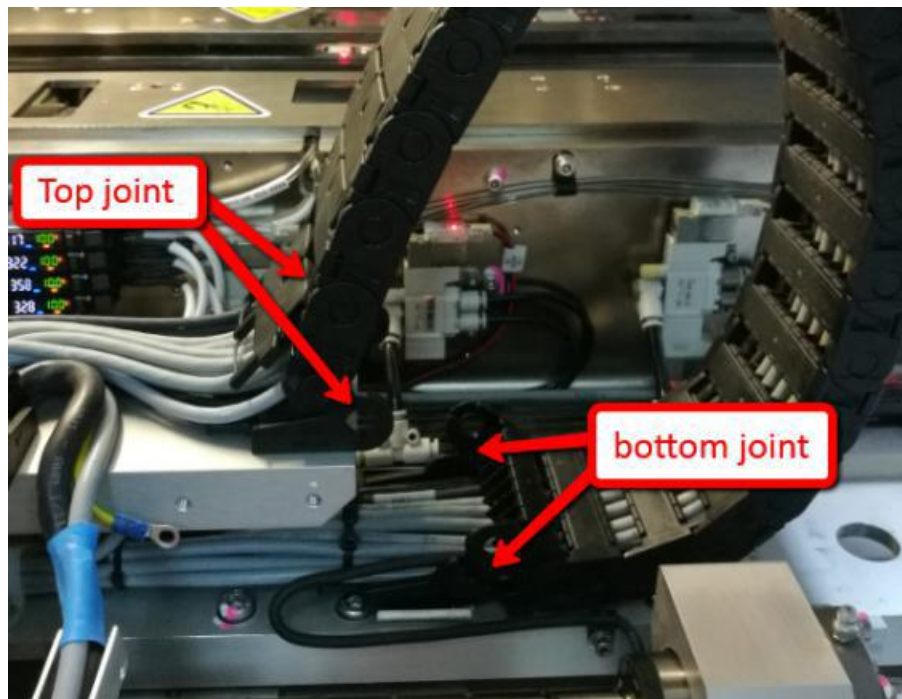


Figure 17

20. Open remaining X-Axis E-chain slot cover and apply cable ties after open each 2-3 slot covers as shown in Figure 18.

Remark: Apply cable ties on non-related cables only.

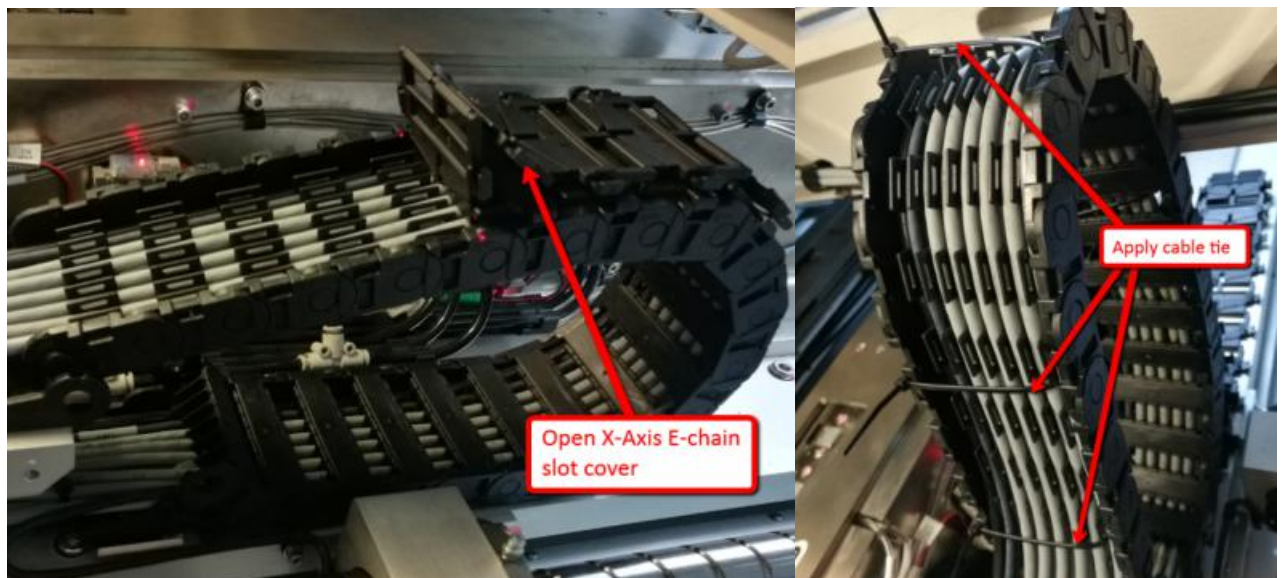


Figure 18

21. Gentle remove X-Axis encoder cable from X-Axis E-chain.

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22. Remove necessary cable tie. Use flat head screw driver to open end joints and slot covers on Y-Axis wide E-chain (right hand side) as shown in Figure 19.

Remark: You may push the stage to rear side to make the removal work easier.

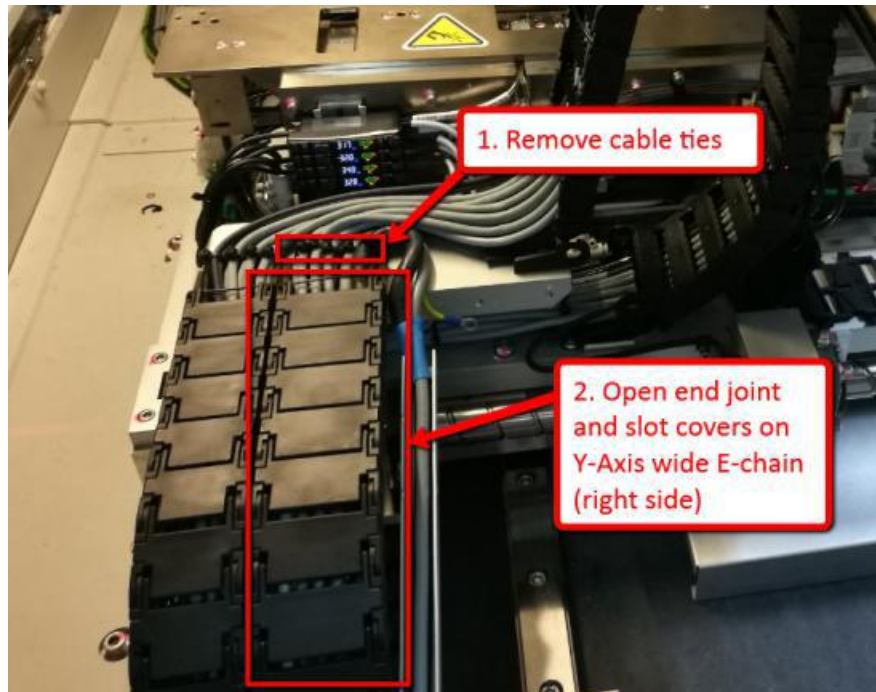


Figure 19

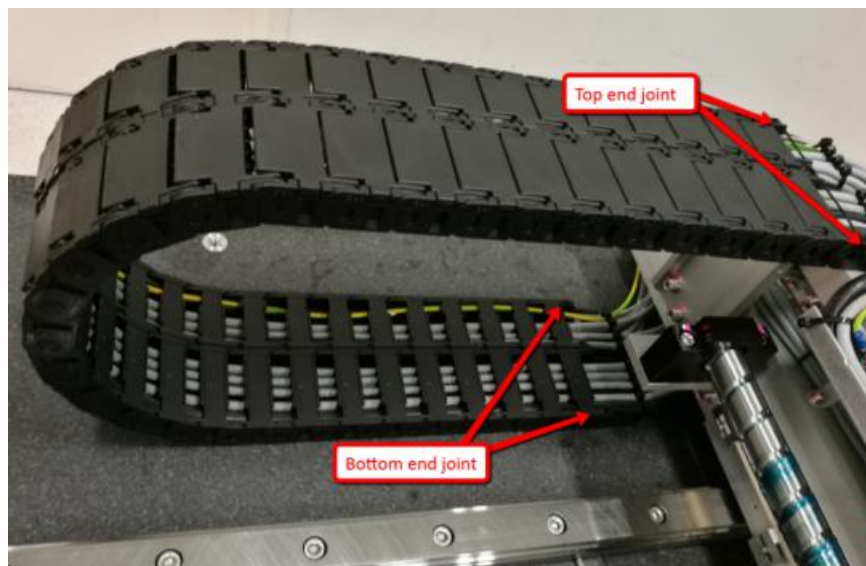


Figure 20

23. Open remaining Y-Axis wide E-chain slot cover and apply cable ties after open 2-3 slot covers.

Remarks: Apply cable ties on non-related cables only.

Remark: E-chain opening and cable removal method are almost same for X-Axis & Y-axis.

24. Gentle remove X-Axis encoder cable from Y-Axis E-chain.
25. Gentle remove X-Axis encoder from machine.

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X-Axis Encoder Replacement

1. Assemble X-Axis Encoder to the bracket and tighten up 2 screws as shown in Figure 21.

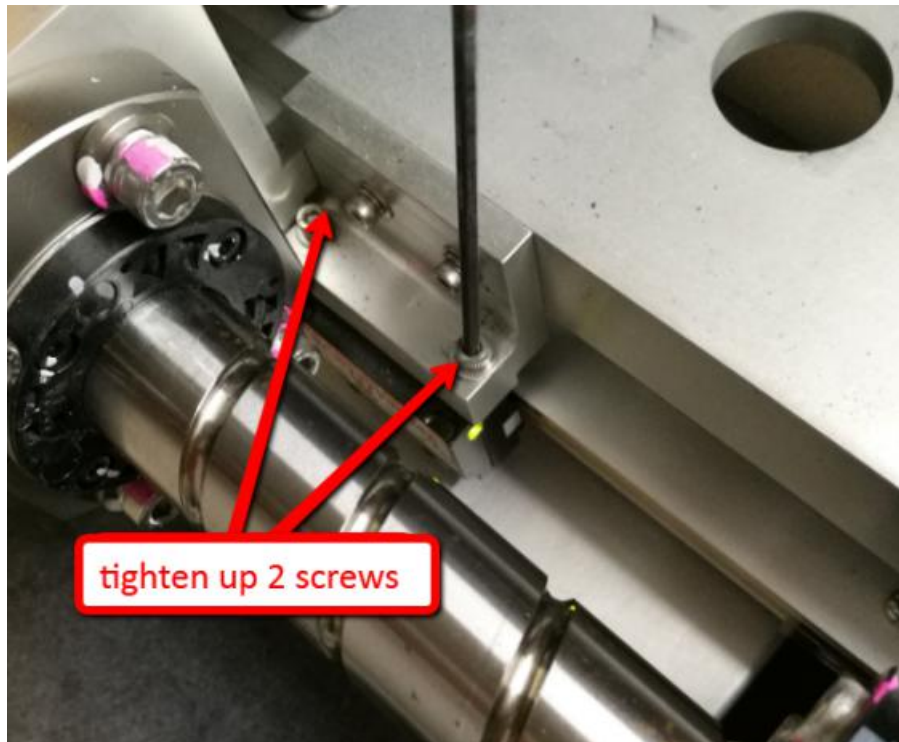


Figure 21

2. Apply cable clamp and tighten up 2 screws as shown in Figure 22.



Figure 22

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- Clamp back X-axis encoder cable and tighten back 2 screws as shown in Figure 23. Apply necessary cable tie.

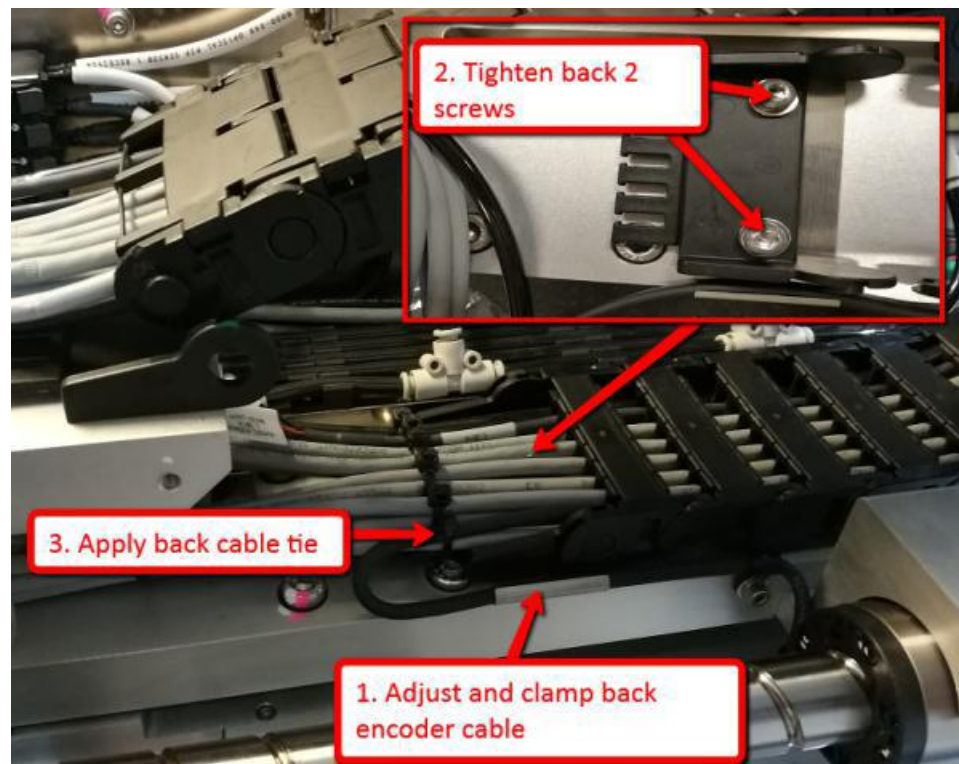


Figure 23

- Gentle route back X-Axis encoder cable and remove necessary cable tie. Close back E-chain slot cover to each partition as shown in Figure 24.

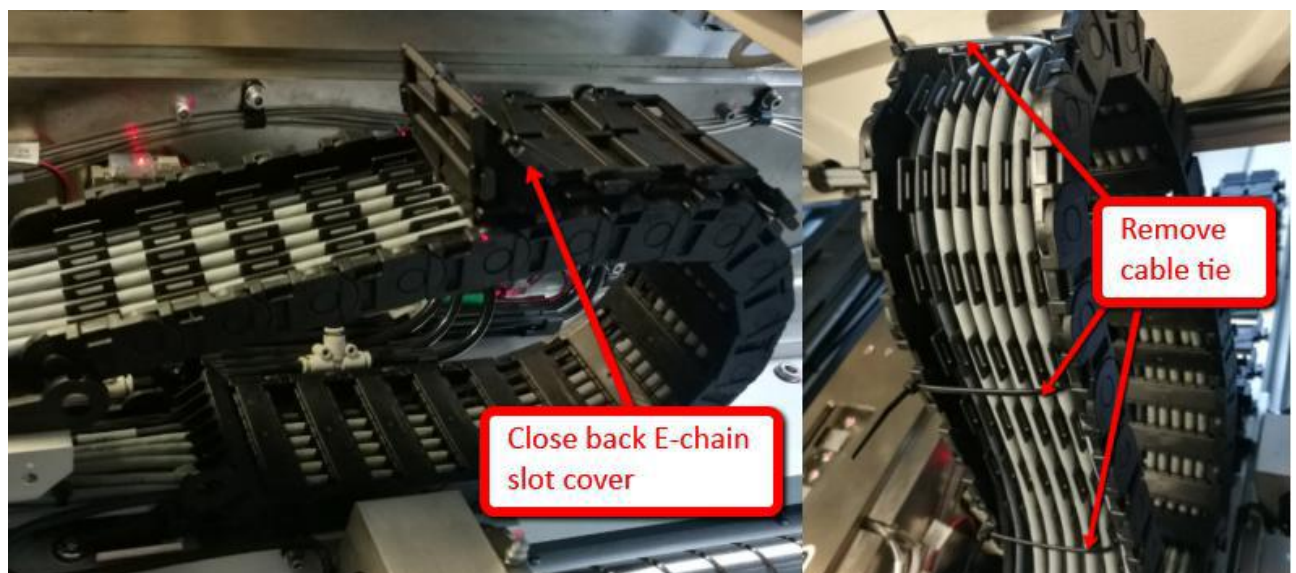


Figure 24

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5. Close back E-chain end joints on both side and apply necessary cable tie as shown in Figure 25.

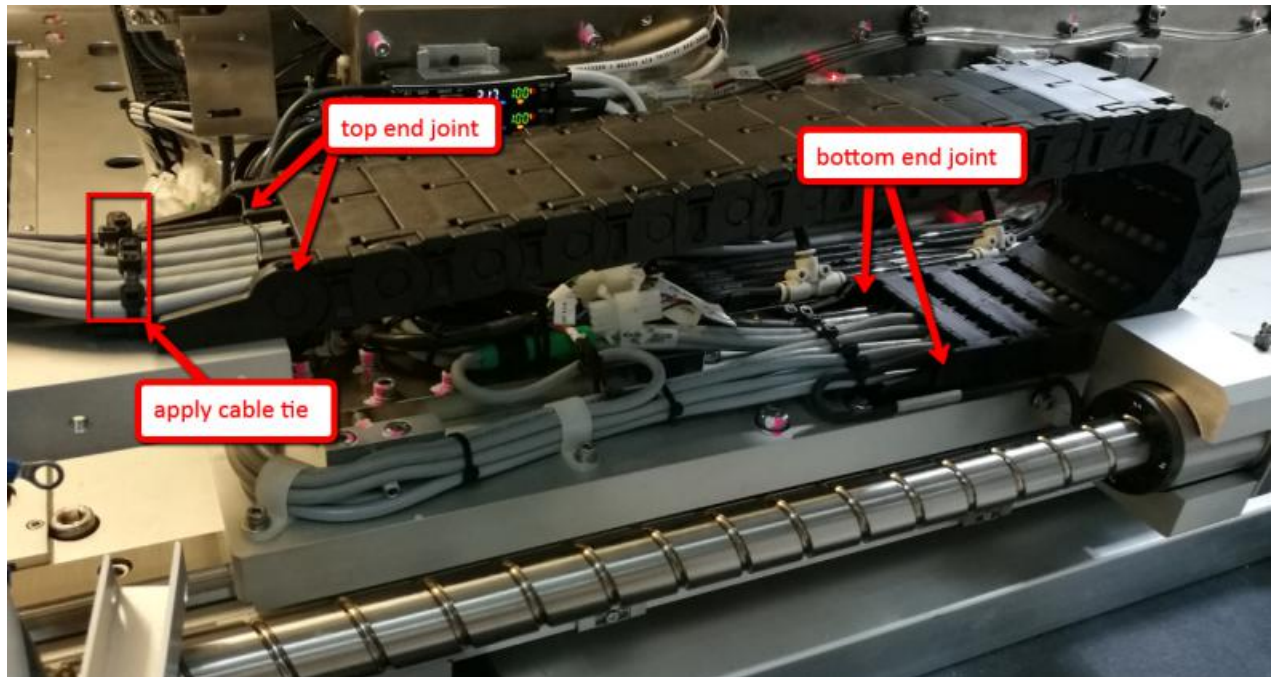


Figure 25

6. Push X-Axis stage to “left and right” for several times as shown in Figure 26. Make sure X-Axis stage is moving smoothly and the cables are well-routed. If not, need to open E-chain and re-route the cables.

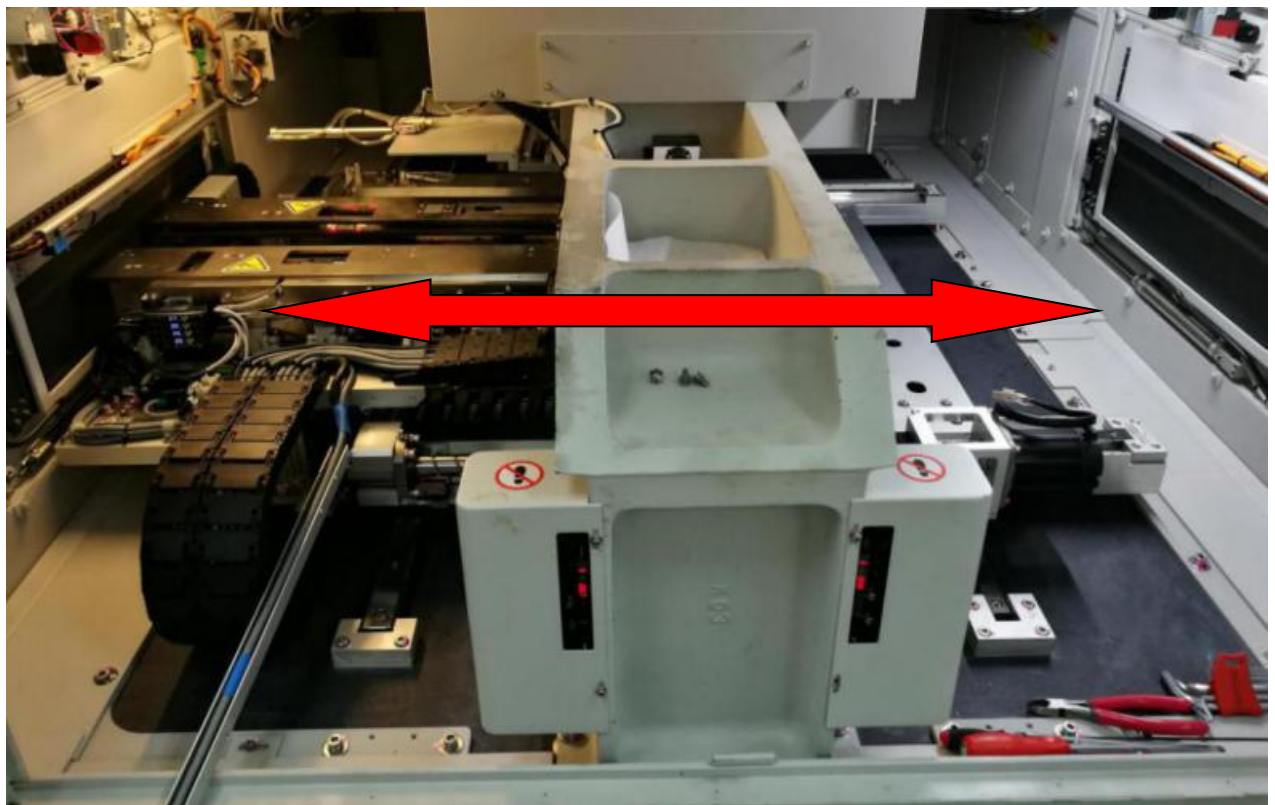


Figure 26

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7. Assemble back X-Axis motor cable bracket and Guard plate. Tighten back 2 screws as shown in Figure 27.

Remark: Grounding wire is screwed together with Guard plate.

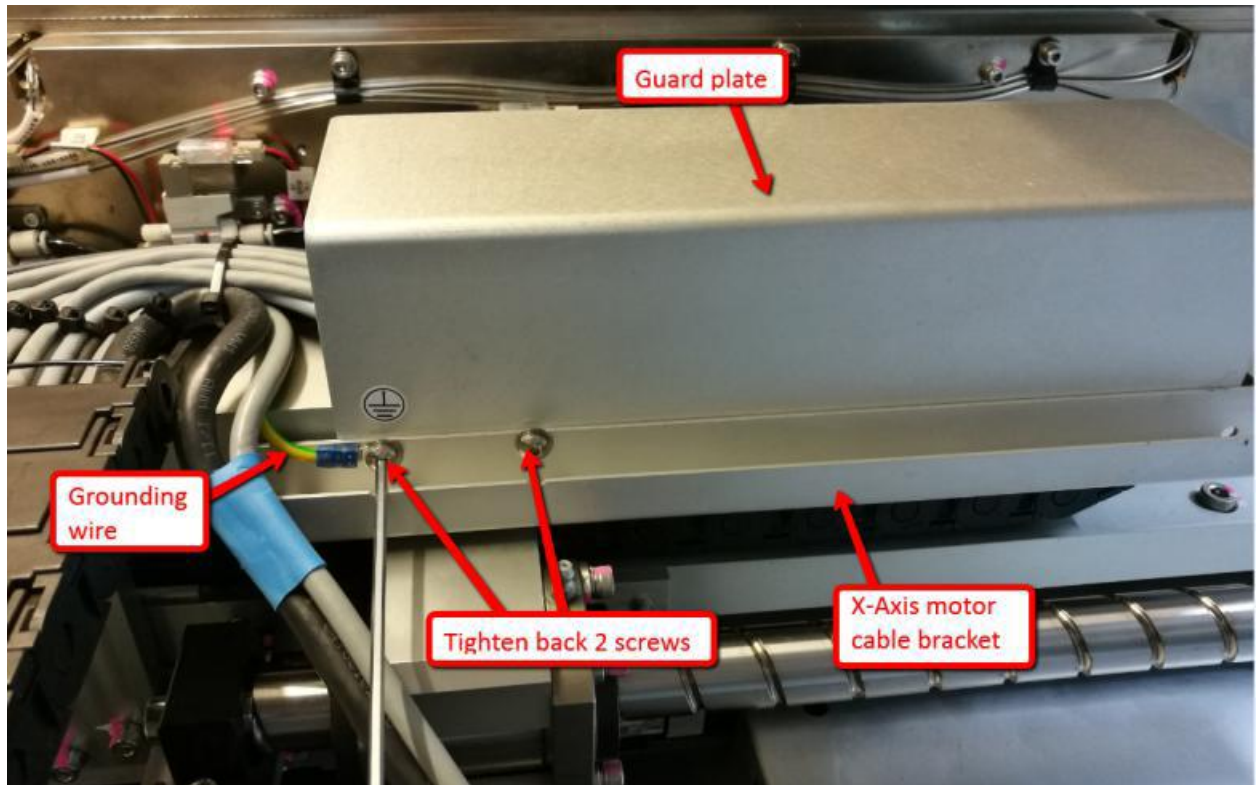


Figure 27

8. Tighten up the screw and plug back X-Axis motor cable connector as shown in Figure 28.

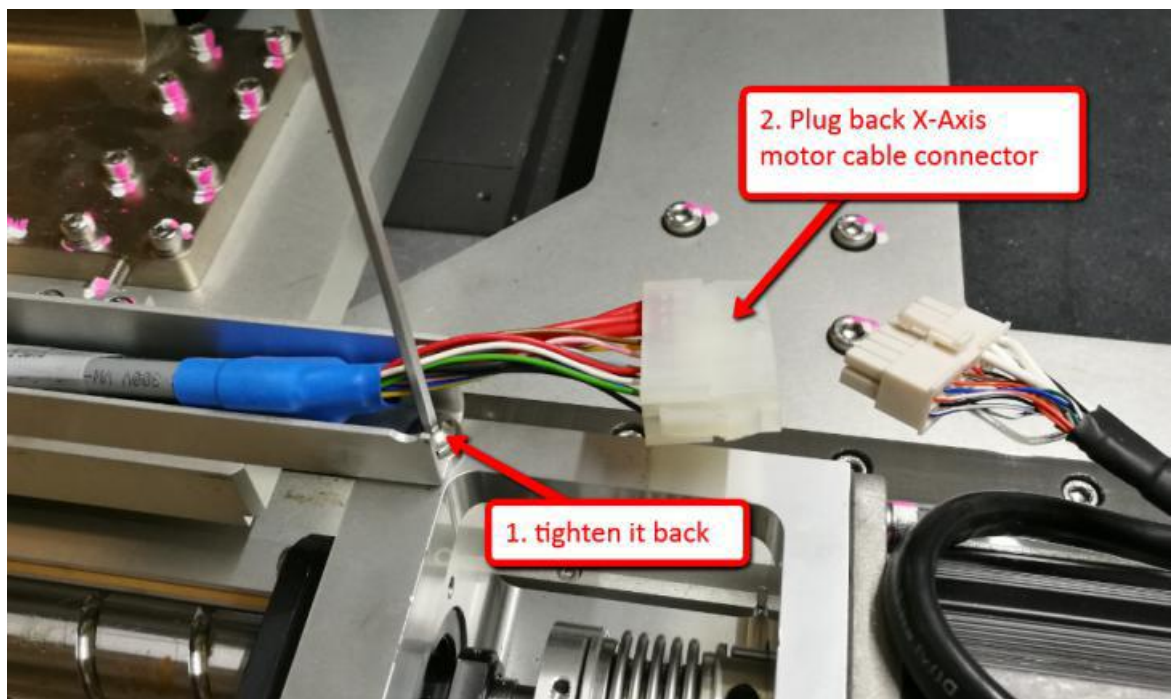


Figure 28

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9. Gentle route back X-Axis encoder cable to Y-Axis E-chain. Remove necessary cable ties and close back E-chain slot covers to each partition as shown in Figure 29.

Remarks: Make sure all cables are not twisted.

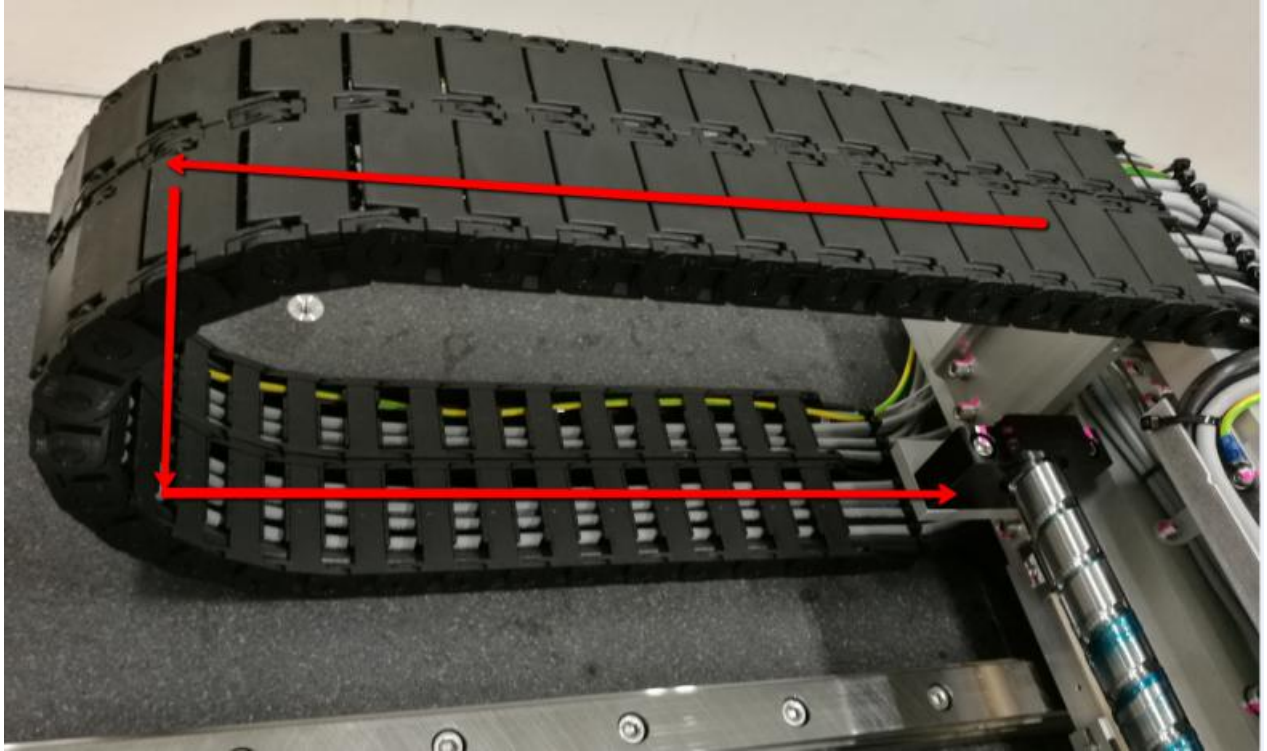


Figure 29

10. Close back end joints on both side for Y-Axis E-chain and apply necessary cable ties as shown in Figure 30.



Figure 30

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11. Move Y-Axis stage to “front and rear” for several times as shown in Figure. Make sure Y-Axis stage is moving smoothly and the cables are well-routed. If not, need to open E-chain and re-route the cables.
12. Route back X-Axis encoder cable to SSCA by following cable routing as shown in Figure 31.
Remark: Apply necessary cable ties.

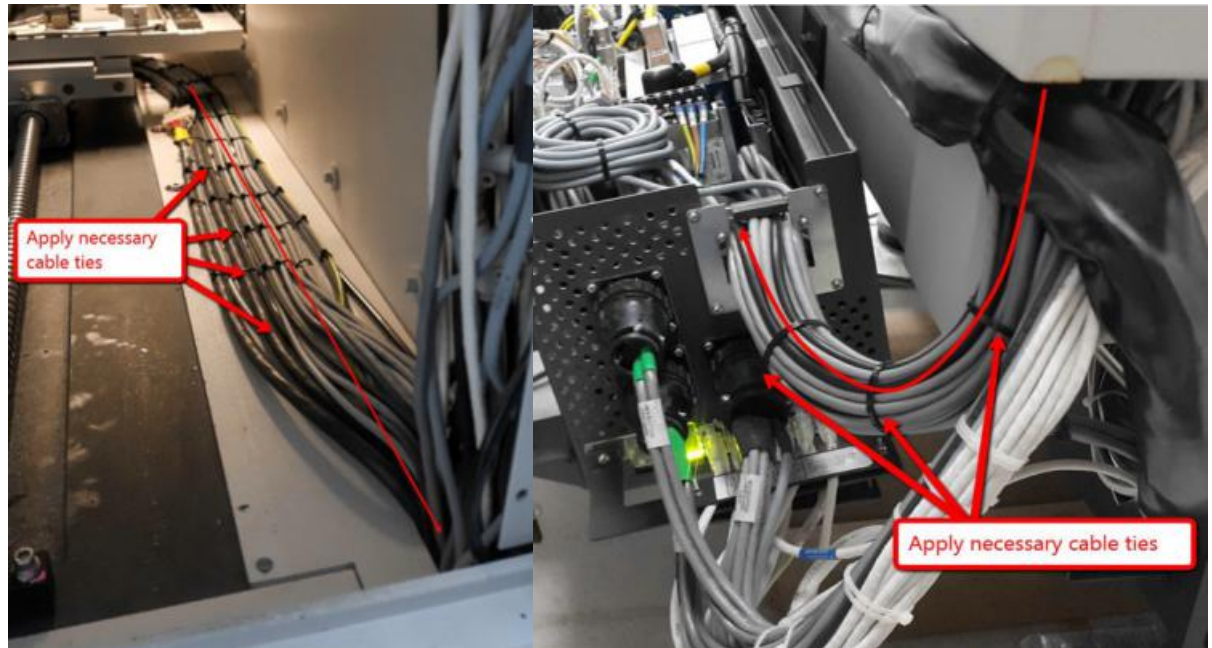


Figure 31

13. Plug X-Axis encoder cable connector back to X-Axis kollmorgen as shown in Figure 32.

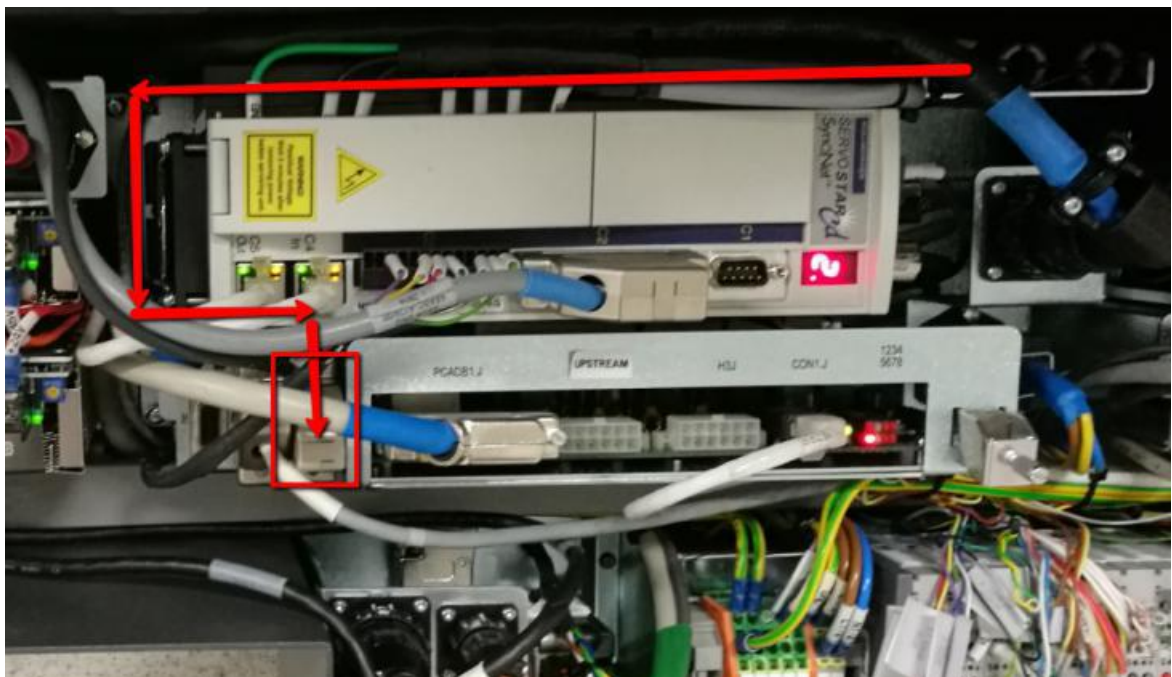


Figure 32

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14. Place cable clamp on its position and tighten back the screw as shown in Figure 33.

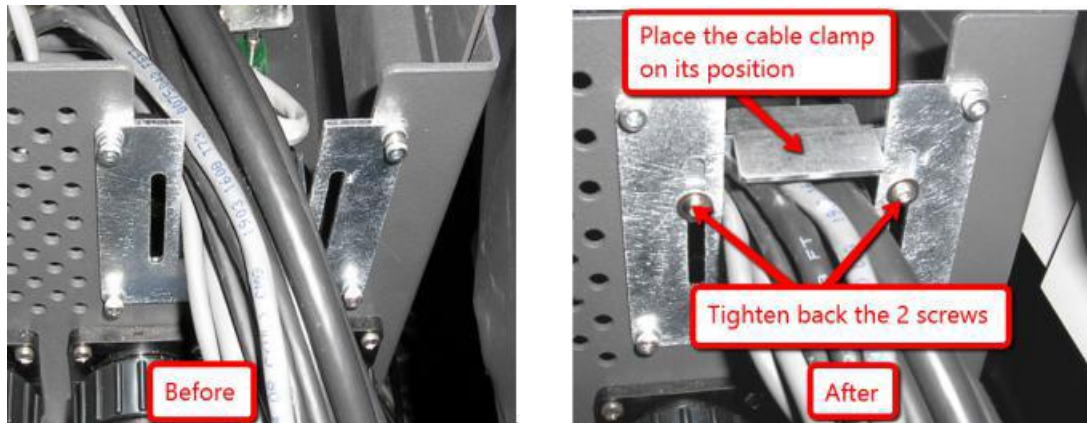


Figure 33

15. Close back lead cover and tighten up 3 screws as shown in Figure 34.

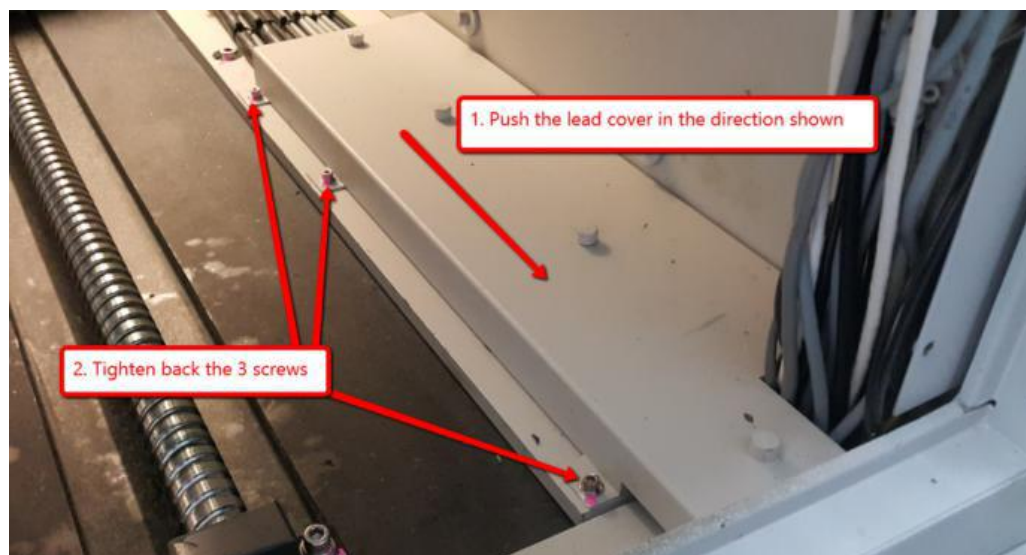


Figure 34

16. Close back lead cover and tighten up 2 screws as shown in Figure 35.



Figure 35

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X-Axis Read Head Encoder Alignment Checking

1. Manually push X-Axis stage from minimum hard stop (Left side) to maximum hard stop (Right side). The encoder should remain green color all the way as shown in Figure 36.

Remark: Encoder will only blink red light when pass through Home Sensor.

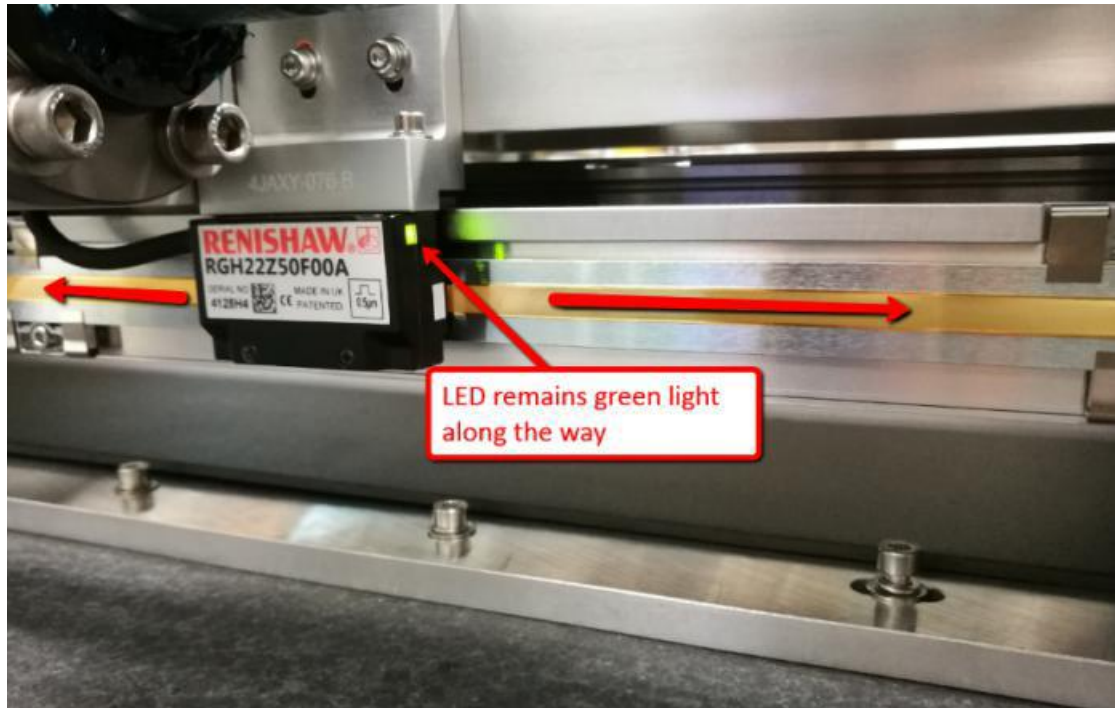


Figure 36

2. If no, continue step 3. If yes, may direct refer to step 7.
3. Loosen 2 screws and place Read Head encoder spacer between encoder and glass scale as shown in Figure 37.

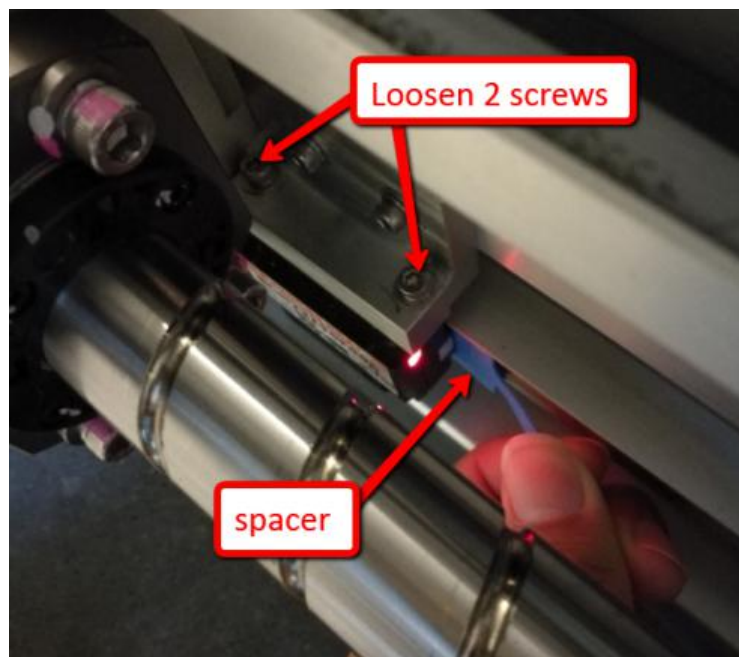


Figure 37

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4. Push encoder toward spacer and tighten back 2 screws as shown in Figure 38. Remove spacer after tighten the screws.

Remark: Spacer still able to pull in and out after tighten.

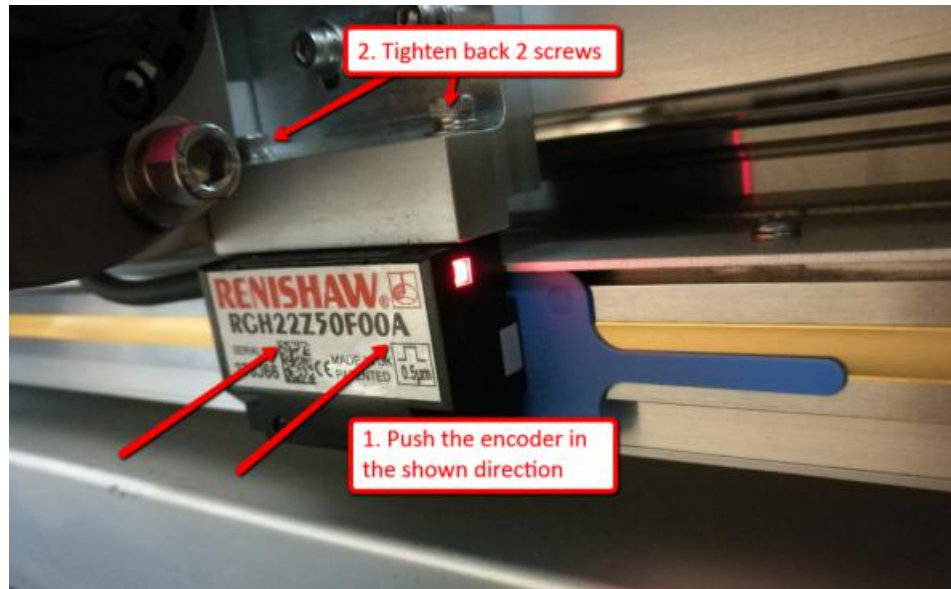


Figure 38

5. Repeat step 1 until encoder remains green light along the X-Axis.
6. Close all door access, including outer barrier.
7. Run 5 loops of long scan.
8. Open X-ray source and perform x-ray survey.
9. Perform 2 loops of CD & A.